

Connecting Arduino to the Internet

If you have just begun working with Arduino, connecting to the internet can be a very daunting task. I remember my own struggles with it in my early days of Arduino coding. With the lack of a proper interface and no inbuilt debugging tools, it can be hard to figure out where you went wrong. In this article, we shall cover everything you need to know about connecting your Arduino board to the internet, including the code.

When I first began playing around with my Arduino Uno, I was very proud of making LEDs blink and sensors beep. But I soon realised that the true potential of an Arduino can only be achieved once you connect it to the internet.

Pretty much any intermediate to advanced level Arduino project will require you to connect to the internet for various reasons, be it logging data on the cloud such as weather data or passing on real-time commands to your device remotely such as via an app on your phone.

In this tutorial article, we are going to look at how to connect your Arduino device to the internet comprehensively. Everything — from the hardware to the circuit and code — will be covered.

Connecting via the Ethernet

The first option for connecting your Arduino to the internet is via an

Ethernet cable. If you are using an Arduino board that comes with a built-in Ethernet port such as an Arduino Yún, then you can skip the ‘Hardware requirements’ section and the circuit design description given below. Just connect the Ethernet cable to your device and start coding. However, if, like most people, you own a simpler version of Arduino such as Arduino Uno that does not have a built-in Ethernet port, then you will have to buy a separate device called ‘Ethernet Shield’ to attach to your Arduino.

Hardware requirements

You will require an Ethernet cable with internet, an Arduino Uno board and an Ethernet Shield. Here’s a description of the circuit design.

1. Connect the Ethernet Shield on top of the Arduino Uno.
2. Connect the Ethernet cable to the Ethernet Shield.

Figure 1 shows the circuit design of Arduino Uno and the Ethernet Shield. After connecting the two, your Arduino Uno should look like the one in Figure 2.

The code

Before getting started with the code for connecting the Arduino to the internet, we first need to include an external library to use in the program. This library will help us establish a connection and send/receive data over the internet.

Note: This library should come preinstalled with your IDE. If for some reason you run into errors, please download this library from the official GitHub repository of the Arduino Project.

```
#include <Ethernet.h>
```

In the next step, we will be defining some constants and variables required

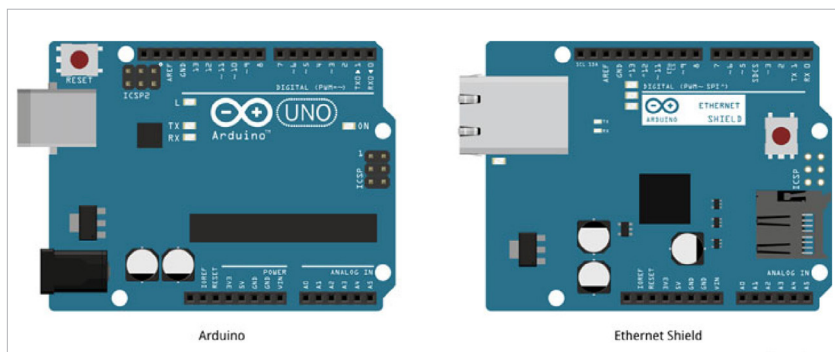


Figure 1: Arduino Uno and Ethernet Shield

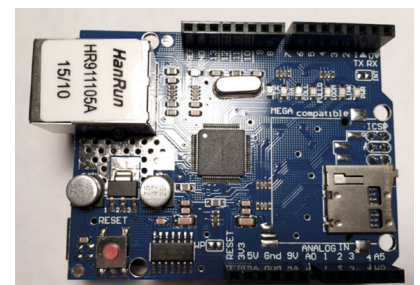


Figure 2: Ethernet Shield attached to Arduino Uno